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OsdagBridge

Open Source Software for Steel Girder Bridge Design

Design Report

Project Name	OsdagBridge Report
Project Location	Roorkee, Uttarakhand
Author / Designer	Aryan Vishala
Reviewer	Nidhikhare12
Organization	FET, GKV
Client Name and Organization	OsdagBridge
Job Number	
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Table of Contents

Executive Summary	2
Project Overview	2
Key Design Outcomes Summary	6
Design Assumptions and Limitations	6
1 Project Information	8
1.1 Project and Design Team Details	8
1.2 Applicable Codes and Standards	8
2 Input Parameters	9
2.1 Basic Inputs (User-Defined)	9
2.2 Additional Inputs	10
3 Loads and Load Combinations	15
4 Analysis Results	18
5 Design Checks	25
5.1 Plate Girder Design	25
5.2 Deck Slab Design	52
5.3 Cross Bracing Design	56
5.4 End Diaphragm Design	61
5.5 Overall Design Check Summary	66
6 Drawings and Visualizations	69
6.1 Bridge Configuration and Layout	69
6.2 Plate Girder — Detailed Views	71
6.3 Cross Bracing Detail	73
6.4 End Diaphragm Detail	74
7 Material Take-off & Quantity Summary	75
8 Standards & Assumptions	77
8.1 Design Standards	77
8.2 Analysis and Design Assumptions of This Version	78
8.3 Known Limitations of This Version	80
References	81

Executive Summary

This section provides a concise summary of the bridge design, key inputs, governing loads, and final design outcomes.

Project Overview

Bridge Type	Highway Bridge
Design Standard	IRC 5, IRC 6, IRC 22, IRC 24, IS 800
Span	40 m
Carriageway Width	20 m
No. of Girders	10
Girder Spacing	2.465 m
Deck Thickness	250.0 mm
Overall Design Status	Fail (Fatigue Normal Stress, Int.Stiff: Leg H_limit, Fatigue Shear Stress)
Governing Check	Int.Stiff: Leg H_limit
Overall Utilization Ratio (max)	1.92 (Girder)



Figure 1 – Overall Bridge Plan



Figure 2 – Typical Cross-Section (with girder, deck, barriers, footpath)

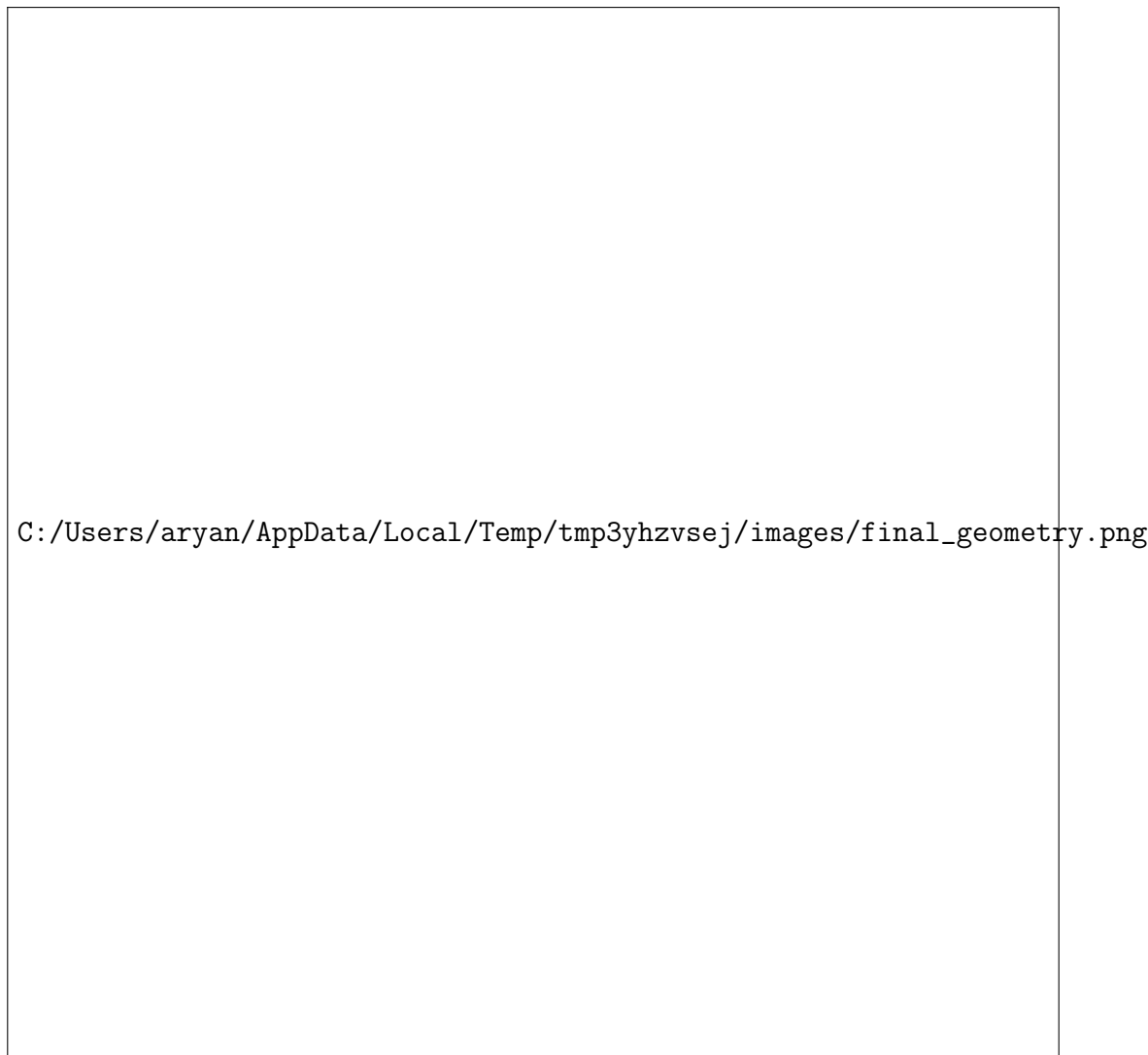


Figure 3 – 3D View of Bridge Superstructure

Table 1 – Final Bridge Geometry (after optimization)

Girder	Member ID	Section Designation	Governing Check	Utilization Ratio
G1	G1M1	2230 x 670 x 28 x 670 x 28	Int.Stiff: Leg H_limit	1.92 (Girder)
G2	G2M1	2230 x 670 x 28 x 670 x 28	Int.Stiff: Leg H_limit	1.92 (Girder)
G3	G3M1	2230 x 670 x 28 x 670 x 28	Int.Stiff: Leg H_limit	1.92 (Girder)
G4	G4M1	2230 x 670 x 28 x 670 x 28	Int.Stiff: Leg H_limit	1.92 (Girder)
G5	G5M1	2230 x 670 x 28 x 670 x 28	Int.Stiff: Leg H_limit	1.92 (Girder)
G6	G6M1	2230 x 670 x 28 x 670 x 28	Int.Stiff: Leg H_limit	1.92 (Girder)
G7	G7M1	2230 x 670 x 28 x 670 x 28	Int.Stiff: Leg H_limit	1.92 (Girder)
G8	G8M1	2230 x 670 x 28 x 670 x 28	Int.Stiff: Leg H_limit	1.92 (Girder)
G9	G9M1	2230 x 670 x 28 x 670 x 28	Int.Stiff: Leg H_limit	1.92 (Girder)
G10	G10M1	2230 x 670 x 28 x 670 x 28	Int.Stiff: Leg H_limit	1.92 (Girder)

Note: Utilization ratio (UR) = demand / capacity. A value < 1.0 indicates a passing check.

Key Design Outcomes Summary

Girder design pass

Cross bracing design pass

End Diaphragm design pass

Deck design pass

Design Assumptions and Limitations

- Additional inputs not provided by the user were assumed by software per IRC/IS code defaults or practical consideration.

- Grillage analysis was performed using OSPGrillage assuming simply supported I-girders.
- Substructure and foundation design are not included in this report.
- Splice connections and bearings are not designed in this version.

1 Project Information

This section records all project metadata as entered by the designer.

1.1 Project and Design Team Details

Project Name	OsdagBridge Report
Project Location	Roorkee, Uttarakhand
Designer	Aryan Vishala
Reviewer	Nidhikhare12
Organization	FET, GKV
Client	OsdagBridge
Software Version	OsdagBridge

1.2 Applicable Codes and Standards

- Indian Roads Congress (IRC) 5: General Features of Design
- Indian Roads Congress (IRC) 6: Loads and Load Combinations
- Indian Roads Congress (IRC) 22: Composite Construction (Limit State Design)
- Indian Roads Congress (IRC) 24: Steel Road Bridges (Limit State Method)
- Indian Roads Congress (IRC) 112: Concrete Road Bridges (deck design)
- Indian Roads Congress Special Publication (IRC SP) 114: Seismic Design of Road Bridges
- Indian Standard (IS) 800: General Construction in Steel
- Indian Standard (IS) 2062: Hot Rolled Structural Steel Specification
- Indian Standard (IS) 6006: Steel Bearings

2 Input Parameters

This section documents all inputs provided to OsdagBridge. User-provided inputs are clearly distinguished from software-assumed defaults. Where the user did not supply a value, the software has applied the IRC/IS code default or an empirical guideline; these are annotated with an asterisk (*).

2.1 Basic Inputs (User-Defined)

Note: These inputs are mandatory and were provided by the user.

Table 2.1: Project Location

Parameter	Value
Project Location	Roorkee, Uttarakhand
Latitude / Longitude	29.8543, 77.888
Seismic Zone (IRC 6)	IV
Basic Wind Speed (IRC 6)	39 m/s
Shade Temp. Max / Min (IRC 6)	47.4 °C / -2.2 °C

Table 2.2: Bridge Geometry

Parameter	Value
Type of Structure	Highway Bridge
Span (m)	40 m
Carriageway Width (m)	20 m
Include Median	No
Footpath	Both Sides
Skew Angle (degrees)	0° (IRC 24 Cl. 504.8 limit: $\pm 15^\circ$)

Table 2.3: Material Selection

Parameter	Value
Girder Steel Grade (IS 2062)	E 350A
Cross Bracing Steel Grade	E 350A
End Diaphragm Steel Grade	E 350A
Concrete Deck Grade (IRC 22)	M40

2.2 Additional Inputs

Where the user has modified additional inputs, those values are reported here. Where no modification was made, the software default is shown.

Table 2.4: Typical Section Details

Parameter	Value
Overall Bridge Width (m)	24.65 m
No. of Girders	10
Girder Spacing (m)	2.465 m
Deck Overhang Width (m)	1.2325 m
Deck Thickness (mm)	250 mm
Footpath Width (m)	1.5 m (IRC 5 Cl. 104.3.6 min: 1.5 m)
No. of Traffic Lanes	5 (per IRC 5 Cl. 104.3.1)

Table 2.5: Components Details

Parameter	Value
Crash Barrier Type	IRC 5 - RCC Crash Barrier
Crash Barrier Load (kN/m)	6.81
Railing Type	IRC 5 - RCC Railing
Railing Load (kN/m)	6.806
Wearing Course Material	Concrete
Wearing Course Thickness (mm)	50.0 mm

Table 2.6: Girder General Information

Girder	Member ID	Design Mode	Girder Type	Girder Symmetry
G1	G1M1	Custom	Welded	Girder Symmetric
G2	G2M1	Custom	Welded	Girder Symmetric
G3	G3M1	Custom	Welded	Girder Symmetric
G4	G4M1	Custom	Welded	Girder Symmetric
G5	G5M1	Custom	Welded	Girder Symmetric
G6	G6M1	Custom	Welded	Girder Symmetric
G7	G7M1	Custom	Welded	Girder Symmetric
G8	G8M1	Custom	Welded	Girder Symmetric
G9	G9M1	Custom	Welded	Girder Symmetric

Girder General Information (continued)

Girder	Member ID	Design Mode	Girder Type	Girder Symmetry
G10	G10M1	Custom	Welded	Girder Symmetric

Table 2.7: Girder Section Dimensions

Girder	Total Depth, D (mm)	Web, t_w (mm)	Top Flange (b_{tf} , t_{tf}) mm	Bottom Flange (b_{bf} , t_{bf}) mm
G1	2230 mm	12 mm	670 mm, 28 mm	670 mm, 28 mm
G2	2230 mm	12 mm	670 mm, 28 mm	670 mm, 28 mm
G3	2230 mm	12 mm	670 mm, 28 mm	670 mm, 28 mm
G4	2230 mm	12 mm	670 mm, 28 mm	670 mm, 28 mm
G5	2230 mm	12 mm	670 mm, 28 mm	670 mm, 28 mm
G6	2230 mm	12 mm	670 mm, 28 mm	670 mm, 28 mm
G7	2230 mm	12 mm	670 mm, 28 mm	670 mm, 28 mm
G8	2230 mm	12 mm	670 mm, 28 mm	670 mm, 28 mm
G9	2230 mm	12 mm	670 mm, 28 mm	670 mm, 28 mm
G10	2230 mm	12 mm	670 mm, 28 mm	670 mm, 28 mm

Table 2.8: Girder Restraint and Stiffener Details

Girder	Torsional / Warping Restraint	Web Philosophy	Intermediate Stiffeners	Longitudinal Stiffeners	Bearing Stiffener
G1	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G2	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G3	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm

Girder Restraint and Stiffener Details (continued)

Girder	Torsional / Warping Restraint	Web Philosophy	Intermediate Stiffeners	Longitudinal Stiffeners	Bearing Stiffener
G4	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G5	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G6	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 3260 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G7	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 3260 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G8	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 3260 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G9	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 3260 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G10	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 3260 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm

Table 2.9: Member Properties: Cross Bracing Details

Location	Member IDs	Type of Bracing	Bracing Section	Spacing (m)
G1 to G2	B1M1 to B1M1	X-Bracing	L 100 x 100x 10	20.0 m

Member Properties: Cross Bracing Details (continued)

Location	Member IDs	Type of Bracing	Bracing Section	Spacing (m)
G2 to G3	B2M1 to B2M1	X-Bracing	IS 100 x 100 x 10	20.0 m
G3 to G4	B3M1 to B3M1	X-Bracing	IS 100 x 100 x 10	20.0 m
G4 to G5	B4M1 to B4M1	X-Bracing	IS 100 x 100 x 10	20.0 m
G5 to G6	B5M1 to B5M1	X-Bracing	IS 100 x 100 x 10	20.0 m
G6 to G7	B6M1 to B6M1	X-Bracing	IS 100 x 100 x 10	20.0 m
G7 to G8	B7M1 to B7M1	X-Bracing	IS 100 x 100 x 10	20.0 m
G8 to G9	B8M1 to B8M1	X-Bracing	IS 100 x 100 x 10	20.0 m
G9 to G10	B9M1 to B9M1	X-Bracing	IS 100 x 100 x 10	20.0 m

Table 2.10: Member Properties: End Diaphragm Details

Location	Member IDs	Type of Bracing	Bracing Section
G1 to G2	E1M1 to E1M2	Cross Bracing	L 100 x 100x 10
G2 to G3	E2M1 to E2M2	Cross Bracing	IS 100 x 100 x 10
G3 to G4	E3M1 to E3M2	Cross Bracing	IS 100 x 100 x 10
G4 to G5	E4M1 to E4M2	Cross Bracing	IS 100 x 100 x 10
G5 to G6	E5M1 to E5M2	Cross Bracing	IS 100 x 100 x 10
G6 to G7	E6M1 to E6M2	Cross Bracing	IS 100 x 100 x 10
G7 to G8	E7M1 to E7M2	Cross Bracing	IS 100 x 100 x 10

Member Properties: End Diaphragm Details (continued)

Location	Member IDs	Type of Bracing	Bracing Section
G8 to G9	E8M1 to E8M2	Cross Bracing	IS 100 x 100 x 10
G9 to G10	E9M1 to E9M2	Cross Bracing	IS 100 x 100 x 10

Table 2.11: Shear Connector Details

Parameter	Value
Stud Diameter (mm)	20.0 mm
Stud Height (mm)	100.0 mm
Stud f_y (MPa)	385.0 MPa
Stud f_u (MPa)	495.0 MPa
No. of Studs per Section	2

Note: All values are per IRC 22 Table 1 unless user-modified.

Table 2.12: Partial Safety Factors

Parameter	Value
γ_{M0} (Yielding / Buckling)	1.1
γ_{M1} (Ultimate Stress)	1.25
γ_C (Concrete, Basic)	1.5
γ_s (Reinforcement)	1.15
γ_v (Shear Connectors)	1.25
γ_{fft} (Fatigue Load)	1.0
γ_{Mft} (Fatigue Strength)	1.35

3 Loads and Load Combinations

This section summarizes all loads applied to the bridge and the load combinations considered for analysis and design.

Table 3.1: Dead Load – Self Weight

Parameter	Value
Steel Self-Weight Applied	76.4 kN/m ³
Concrete Deck Weight	24.5 kN/m ³
Self-Weight Factor	1

Table 3.2: Dead Load for Surfacing (DW)

Parameter	Value
Wearing Course Load	Concrete x 50 mm
Additional SIDL (Crash Barrier)	6.81 kN/m per barrier
Railing Load	6.806 kN/m

Table 3.3: Vehicle Live Loads

Parameter	Value
Vehicles Considered	Class A, Class 70R (Wheeled), Class 70R (Tracked), Class AA (Wheeled), Class AA (Tracked), Class SV, Class 70R (Bogie), Class Fatigue
Impact Factor (IRC 6)	Class A: 1.168, Class 70R (Wheeled): 1.100
Braking Load (IRC 6)	N/A
Centrifugal Force	N/A

Table 3.4: Footway Load

Parameter	Value
Load Type	IRC 6 Cl. 206.1
Intensity	4.905 kN/m ²

Table 3.5: Wind Load (WL) — per IRC 6

Parameter	Value
Basic Wind Speed, V _b	39 m/s [from Project Location]
Terrain Type	Plain Terrain

Wind Load (WL) — per IRC 6 (continued)

Parameter	Value
Average Exposed Height, H (m)	10 m
Hourly Mean Wind Speed, Vz	32.85 m/s
Hourly Wind Pressure, Pz	647.6 N/m ²
Transverse Wind Force	N/A
Longitudinal Wind Force	N/A
Vertical Wind Force	N/A

Table 3.6: Earthquake Load (EL) — per IRC 6

Parameter	Value
Seismic Zone	IV [from Project Location]
Zone Factor, Z	0.24
Importance Factor, I	1
Type of Soil	Type I – Rocky or Hard
Sa/g	2.8
Horizontal Seismic Coefficient, Ah	0.336
Vertical Seismic Coefficient, Av	0.224
Horizontal Seismic Force (longitudinal)	N/A
Horizontal Seismic Force (transverse)	N/A

Table 3.7: Temperature Load (TL) — per IRC 6

Parameter	Value
Maximum Shade Temperature	47.4 °C
Minimum Shade Temperature	-2.2 °C
Effective Bridge Temp. Range	-12.2 °C to 62.4 °C
Temperature Rise / Fall for Design	+37.3 °C / -37.3 °C

Table 3.8: Load Combinations

Combination ID	Load Cases
ULS-01	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.50) + WL(0.90) + TL(0.90)
ULS-02	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.15) + WL(1.50) + TL(0.90)
ULS-03	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.15) + WL(0.90) + TL(1.50)
ULS-04	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + VC(1.00)
ULS-05	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + BI(1.00)
ULS-06	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + FB(1.00)
ULS-07	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(0.20) + TL(0.50) + EQ(1.50)
ULS-08	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(0.20) + TL(0.50) + EQ(0.75)
SLS-01	DL(1.00) + SIDL(1.20/1.00) + LL(1.00) + WL(0.60) + TL(0.60)
SLS-02	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(1.00) + TL(0.60)
SLS-03	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(0.60) + TL(1.00)
SLS-04	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(0.50) + TL(0.50)
SLS-05	DL(1.00) + SIDL(1.20/1.00) + LL(0.20) + WL(0.60) + TL(0.50)
SLS-06	DL(1.00) + SIDL(1.20/1.00) + LL(0.20) + WL(0.50) + TL(0.60)
SLS-07	DL(1.00) + SIDL(1.20/1.00) + LL(0.00) + WL(0.00) + TL(0.50)

Note: All IRC 6 load combinations are auto-generated by OsdagBridge. User-defined custom combinations, if any, are appended.

4 Analysis Results

A grillage model was used for structural analysis. The deck is idealized as a grid of elastic beam elements — longitudinal members represent the composite steel girders with effective slab, and transverse members represent the slab or cross frames. This section summarizes the critical output from that analysis.

Table 4.1: Summary of Maximum Demands

Load Case/ Comb.	Bending Moment			Shear Force			Reaction at Supports	
	Max (kNm)	Girder	Loc. (m)	Max (kN)	Girder	Loc. (m)	Left	Right
Envelope SLS	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
SEISMIC_1: 1.35DL + 1.75DW + 0.2LL + 1.5EL	4357.612	G5	17.778	613.227	G1	0.000	-5096.764	3878.421
SEISMIC_2: 1.35DL + 1.75DW + 0.2LL + 0.75EL	4357.612	G5	17.778	613.227	G1	0.000	-5096.764	3878.421
SLS_FREQUENT_ 5: 1.0DL + 1.0DW + 0.2LL + 0.6WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
SLS_QP_1: 1.0DL + 1.2DW	4124.243	G5	19.259	545.121	G1	0.000	-4247.465	4247.170
1.5 EQ (c)	1387.033	G5	19.259	178.387	G1	0.000	1426.815	-1426.815
SW	801.948	G5	19.259	102.387	G1	0.000	-825.221	825.221
1.0 DL + 1.0 LL	4764.316	G5	17.778	665.264	G1	0.000	-5517.960	4299.617
EQ_Z	0.000	—	—	0.000	—	—	0.000	-0.000
BASIC_6: 1.0DL + 1.0DW + 1.15LL + 0.9WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
1.0 DW	513.721	G5	20.741	66.888	G1	38.519	-530.013	530.013
ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	5272.711	G5	17.778	732.152	G1	0.000	-6047.974	4829.631
Crash barrier load	148.208	G1	20.741	23.044	G1	38.519	-142.198	147.401
BASIC_1: 1.35DL + 1.75DW + 1.5LL + 0.9WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
BASIC_3: 1.35DL + 1.75DW + 1.15LL + 0.9WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774

Load Case/ Comb.	Bending Moment			Shear Force			Reaction at Supports	
	Max (kNm)	Girder	Loc. (m)	Max (kN)	Girder	Loc. (m)	Left	Right
SLS_RARE_2: 1.0DL + 1.2DW + 0.75LL + 1.0WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
BASIC_4: 1.0DL + 1.0DW + 1.5LL + 0.9WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
EQ_Y	924.689	G5	19.259	118.925	G1	0.000	951.210	-951.210
1.0 DL	3610.522	G5	19.259	478.233	G1	0.000	-3717.452	3717.156
DD	2568.605	G5	19.259	334.440	G1	0.000	-2650.067	2650.067
SLS_FREQUENT_3: 1.0DL + 1.2DW + 0.2LL + 0.5WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
EQ_X	0.000	—	—	0.000	—	—	0.000	-0.000
BASIC_5: 1.0DL + 1.0DW + 1.15LL + 1.5WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
SLS_FREQUENT_2: 1.0DL + 1.2DW + 0.2LL + 0.6WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
ACCIDENTAL_2: 1.0DL + 1.0DW + 0.75LL	5272.711	G5	17.778	732.152	G1	0.000	-6047.974	4829.631
ACCIDENTAL_3: 1.0DL + 1.0DW + 0.75LL	5272.711	G5	17.778	732.152	G1	0.000	-6047.974	4829.631
SLS_FREQUENT_6: 1.0DL + 1.0DW + 0.2LL + 0.5WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
Footpath load	248.832	G1	19.259	41.405	G1	0.000	-242.163	241.868
1.0 WL	285.663	G5	20.741	36.739	G1	38.519	293.856	-293.856
SLS_RARE_4: 1.0DL + 1.0DW + 1.0LL + 0.6WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
WL Uplift	285.663	G5	19.259	36.739	G1	0.000	293.856	-293.856
WL Transverse	0.000	—	—	0.000	—	—	-0.000	0.000

Load Case/ Comb.	Bending Moment			Shear Force			Reaction at Supports	
	Max (kNm)	Girder	Loc. (m)	Max (kN)	Girder	Loc. (m)	Left	Right
SLS_RARE_1: 1.0DL + 1.2DW + 1.0LL + 0.6WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
WL Longitudinal	0.000	—	—	0.000	—	—	-0.000	0.000
SLS_RARE_5: 1.0DL + 1.0DW + 0.75LL + 1.0WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
SEISMIC_4: 1.0DL + 1.0DW + 0.2LL + 0.75EL	4357.612	G5	17.778	613.227	G1	0.000	-5096.764	3878.421
SLS_QP_2: 1.0DL + 1.0DW	4124.243	G5	19.259	545.121	G1	0.000	-4247.465	4247.170
SLS_FREQUENT_4: 1.0DL + 1.0DW + 0.75LL + 0.5WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
1.5 EQ (b)	416.110	G5	20.741	53.516	G1	38.519	428.044	-428.044
SEISMIC_3: 1.0DL + 1.0DW + 0.2LL + 1.5EL	4357.612	G5	17.778	613.227	G1	0.000	-5096.764	3878.421
SLS_RARE_3: 1.0DL + 1.2DW + 0.75LL + 0.6WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
Envelope ULS	5272.711	G5	17.778	732.152	G1	0.000	-6047.974	4829.631
1.5 EQ (a)	416.110	G5	20.741	53.516	G1	38.519	428.044	-428.044
SLS_RARE_6: 1.0DL + 1.0DW + 0.75LL + 0.6WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
BASIC_2: 1.35DL + 1.75DW + 1.15LL + 1.5WL	4990.011	G5	17.778	695.412	G1	0.000	-5754.117	4535.774
Railing load	33.656	G1	20.741	5.988	G1	38.519	-31.227	32.394

Table 4.2: Reactions at Supports

Load Case	Left Support (kN)	Right Support (kN)
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Table 4.3: Deflection Summary (Live Load & Total Load)

parameter	value
Deflection due to Live Load, δ_{LL}	17.298 mm
Allowable Live Load Deflection (Δ_{allow})	$L/800 = 50.0$ mm
Live Load Deflection Check Status	PASS
Deflection due to Total Load, δ_{total}	72.433 mm
Allowable Total Deflection (Δ_{allow})	$L/600 = 66.7$ mm
Total Load Deflection Check Status	FAIL



Bending Moment Envelope (Envelope ULS): Max/min BM along span. X-axis: distance from left support (m). Y-axis: Bending Moment (kN-m).



Shear Force Envelope (Envelope ULS): Max/min SF along span. X-axis: distance from left support (m). Y-axis: Shear Force (kN).



Vertical Deflection D_y (1.0 LL): Maximum deflection along span. Load Case: 1.0 LL, Combination: D_y . Nodes shown. Isometric view.

5 Design Checks

This section presents all structural design checks performed by OsdagBridge. For each member, the demand from the governing load combination, the code-based capacity, and the utilization ratio are tabulated. All checks reference IS 800:2007 and IRC 22:2014 unless stated otherwise.

5.1 Plate Girder Design

Table 5.1: Girder Section Properties (Final Optimized / User-selected)

Girder	Parameter	Value
G1	Depth, D (mm)	2230
	Top Flange Width, b_f (mm)	670
	Bottom Flange Width, b_f (mm)	670
	Top Flange Thickness, t_f (mm)	28
	Bottom Flange Thickness, t_f (mm)	28
	Web Thickness, t_w (mm)	12
	Gross Area, A (cm^2)	636.08
	Moment of Inertia, I_z (cm^4)	5,575,915.69
	Elastic Section Modulus, Z_{ez} (cm^3)	50,008.21
	Plastic Section Modulus, Z_{pz} (cm^3)	55,488.35
	Effective Slab Width, b_{eff} (mm)	2465
	Transformed Composite I_z (cm^4)	11,131,466.22
	Depth to Plastic Neutral Axis (mm)	276.8
G2	Depth, D (mm)	2230
	Top Flange Width, b_f (mm)	670
	Bottom Flange Width, b_f (mm)	670
	Top Flange Thickness, t_f (mm)	28
	Bottom Flange Thickness, t_f (mm)	28
	Web Thickness, t_w (mm)	12
	Gross Area, A (cm^2)	636.08
	Moment of Inertia, I_z (cm^4)	5,575,915.69
	Elastic Section Modulus, Z_{ez} (cm^3)	50,008.21
	Plastic Section Modulus, Z_{pz} (cm^3)	55,488.35
	Effective Slab Width, b_{eff} (mm)	2465
	Transformed Composite I_z (cm^4)	11,131,466.22
	Depth to Plastic Neutral Axis (mm)	276.8

Girder Section Properties (Final Optimized / User-selected) (continued)

Girder	Parameter	Value
G3	Depth, D (mm)	2230
	Top Flange Width, b_f (mm)	670
	Bottom Flange Width, b_f (mm)	670
	Top Flange Thickness, t_f (mm)	28
	Bottom Flange Thickness, t_f (mm)	28
	Web Thickness, t_w (mm)	12
	Gross Area, A (cm^2)	636.08
	Moment of Inertia, I_z (cm^4)	5,575,915.69
	Elastic Section Modulus, Z_{ez} (cm^3)	50,008.21
	Plastic Section Modulus, Z_{pz} (cm^3)	55,488.35
	Effective Slab Width, b_{eff} (mm)	2465
	Transformed Composite I_z (cm^4)	11,131,466.22
	Depth to Plastic Neutral Axis (mm)	276.8
G4	Depth, D (mm)	2230
	Top Flange Width, b_f (mm)	670
	Bottom Flange Width, b_f (mm)	670
	Top Flange Thickness, t_f (mm)	28
	Bottom Flange Thickness, t_f (mm)	28
	Web Thickness, t_w (mm)	12
	Gross Area, A (cm^2)	636.08
	Moment of Inertia, I_z (cm^4)	5,575,915.69
	Elastic Section Modulus, Z_{ez} (cm^3)	50,008.21
	Plastic Section Modulus, Z_{pz} (cm^3)	55,488.35
	Effective Slab Width, b_{eff} (mm)	2465
	Transformed Composite I_z (cm^4)	11,131,466.22
	Depth to Plastic Neutral Axis (mm)	276.8
G5	Depth, D (mm)	2230
	Top Flange Width, b_f (mm)	670
	Bottom Flange Width, b_f (mm)	670
	Top Flange Thickness, t_f (mm)	28
	Bottom Flange Thickness, t_f (mm)	28
	Web Thickness, t_w (mm)	12
	Gross Area, A (cm^2)	636.08
	Moment of Inertia, I_z (cm^4)	5,575,915.69

Girder Section Properties (Final Optimized / User-selected) (continued)

Girder	Parameter	Value
	Elastic Section Modulus, Z_{ez} (cm^3)	50,008.21
	Plastic Section Modulus, Z_{pz} (cm^3)	55,488.35
	Effective Slab Width, b_{eff} (mm)	2465
	Transformed Composite I_z (cm^4)	11,131,466.22
	Depth to Plastic Neutral Axis (mm)	276.8
G6	Depth, D (mm)	2230
	Top Flange Width, b_f (mm)	670
	Bottom Flange Width, b_f (mm)	670
	Top Flange Thickness, t_f (mm)	28
	Bottom Flange Thickness, t_f (mm)	28
	Web Thickness, t_w (mm)	12
	Gross Area, A (cm^2)	636.08
	Moment of Inertia, I_z (cm^4)	5,575,915.69
	Elastic Section Modulus, Z_{ez} (cm^3)	50,008.21
	Plastic Section Modulus, Z_{pz} (cm^3)	55,488.35
	Effective Slab Width, b_{eff} (mm)	2465
	Transformed Composite I_z (cm^4)	11,131,466.22
	Depth to Plastic Neutral Axis (mm)	276.8
G7	Depth, D (mm)	2230
	Top Flange Width, b_f (mm)	670
	Bottom Flange Width, b_f (mm)	670
	Top Flange Thickness, t_f (mm)	28
	Bottom Flange Thickness, t_f (mm)	28
	Web Thickness, t_w (mm)	12
	Gross Area, A (cm^2)	636.08
	Moment of Inertia, I_z (cm^4)	5,575,915.69
	Elastic Section Modulus, Z_{ez} (cm^3)	50,008.21
	Plastic Section Modulus, Z_{pz} (cm^3)	55,488.35
	Effective Slab Width, b_{eff} (mm)	2465
	Transformed Composite I_z (cm^4)	11,131,466.22
	Depth to Plastic Neutral Axis (mm)	276.8
G8	Depth, D (mm)	2230
	Top Flange Width, b_f (mm)	670
	Bottom Flange Width, b_f (mm)	670

Girder Section Properties (Final Optimized / User-selected) (continued)

Girder	Parameter	Value
	Top Flange Thickness, t_f (mm)	28
	Bottom Flange Thickness, t_f (mm)	28
	Web Thickness, t_w (mm)	12
	Gross Area, A (cm^2)	636.08
	Moment of Inertia, I_z (cm^4)	5,575,915.69
	Elastic Section Modulus, Z_{ez} (cm^3)	50,008.21
	Plastic Section Modulus, Z_{pz} (cm^3)	55,488.35
	Effective Slab Width, b_{eff} (mm)	2465
	Transformed Composite I_z (cm^4)	11,131,466.22
	Depth to Plastic Neutral Axis (mm)	276.8
G9	Depth, D (mm)	2230
	Top Flange Width, b_f (mm)	670
	Bottom Flange Width, b_f (mm)	670
	Top Flange Thickness, t_f (mm)	28
	Bottom Flange Thickness, t_f (mm)	28
	Web Thickness, t_w (mm)	12
	Gross Area, A (cm^2)	636.08
	Moment of Inertia, I_z (cm^4)	5,575,915.69
	Elastic Section Modulus, Z_{ez} (cm^3)	50,008.21
	Plastic Section Modulus, Z_{pz} (cm^3)	55,488.35
	Effective Slab Width, b_{eff} (mm)	2465
	Transformed Composite I_z (cm^4)	11,131,466.22
	Depth to Plastic Neutral Axis (mm)	276.8
G10	Depth, D (mm)	2230
	Top Flange Width, b_f (mm)	670
	Bottom Flange Width, b_f (mm)	670
	Top Flange Thickness, t_f (mm)	28
	Bottom Flange Thickness, t_f (mm)	28
	Web Thickness, t_w (mm)	12
	Gross Area, A (cm^2)	636.08
	Moment of Inertia, I_z (cm^4)	5,575,915.69
	Elastic Section Modulus, Z_{ez} (cm^3)	50,008.21
	Plastic Section Modulus, Z_{pz} (cm^3)	55,488.35
	Effective Slab Width, b_{eff} (mm)	2465

Girder Section Properties (Final Optimized / User-selected) (continued)

Girder	Parameter	Value
	Transformed Composite I_z (cm^4)	11,131,466.22
	Depth to Plastic Neutral Axis (mm)	276.8

Table 5.2: Girder Section Classification

Girder	Element	Slenderness Ratio	Class Limit	Classification
G1	Top Flange	11.75	11.49	Slender
	Web	181.17	106.49	Slender
	Overall Section	—	—	Slender
G2	Top Flange	11.75	11.49	Slender
	Web	181.17	106.49	Slender
	Overall Section	—	—	Slender
G3	Top Flange	11.75	11.49	Slender
	Web	181.17	106.49	Slender
	Overall Section	—	—	Slender
G4	Top Flange	11.75	11.49	Slender
	Web	181.17	106.49	Slender
	Overall Section	—	—	Slender
G5	Top Flange	11.75	11.49	Slender
	Web	181.17	106.49	Slender
	Overall Section	—	—	Slender
G6	Top Flange	11.75	11.49	Slender
	Web	181.17	106.49	Slender
	Overall Section	—	—	Slender
G7	Top Flange	11.75	11.49	Slender
	Web	181.17	106.49	Slender
	Overall Section	—	—	Slender
G8	Top Flange	11.75	11.49	Slender
	Web	181.17	106.49	Slender
	Overall Section	—	—	Slender
G9	Top Flange	11.75	11.49	Slender
	Web	181.17	106.49	Slender
	Overall Section	—	—	Slender

Girder Section Classification (continued)

Girder	Element	Slenderness Ratio	Class Limit	Classification
G10	Top Flange	11.75	11.49	Slender
	Web	181.17	106.49	Slender
	Overall Section	—	—	Slender

Note: IS 800:2007 Table 2

Table 5.3: Moment Capacity Check

Girder	Parameter	Reference	Value	Status
G1	Applied Moment, M_u	Governing LC (ULS)	5272.71 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	23,734.4 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.22	PASS
G2	Applied Moment, M_u	Governing LC (ULS)	5272.71 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	23,734.4 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.22	PASS
G3	Applied Moment, M_u	Governing LC (ULS)	5272.71 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	23,734.4 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.22	PASS
G4	Applied Moment, M_u	Governing LC (ULS)	5272.71 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	23,734.4 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.22	PASS
G5	Applied Moment, M_u	Governing LC (ULS)	5272.71 kN-m	—

Moment Capacity Check (continued)

Girder	Parameter	Reference	Value	Status
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	23,734.4 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.22	PASS
G6	Applied Moment, M_u	Governing LC (ULS)	5272.71 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	23,734.4 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.22	PASS
G7	Applied Moment, M_u	Governing LC (ULS)	5272.71 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	23,734.4 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.22	PASS
G8	Applied Moment, M_u	Governing LC (ULS)	5272.71 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	23,734.4 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.22	PASS
G9	Applied Moment, M_u	Governing LC (ULS)	5272.71 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	23,734.4 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.22	PASS
G10	Applied Moment, M_u	Governing LC (ULS)	5272.71 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	23,734.4 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.22	PASS

Note: IRC 22 Cl. 603.3.1, IS 800 Cl. 8.2.1

Table 5.4: Shear Capacity Check

Girder	Parameter	Reference	Value	Status
G1	Applied Shear, V_u	Governing LC (ULS)	603.36 kN	—
	Shear Area, A_v	$d_w \times t_w$	26,088 mm ²	—
	Panel Aspect Ratio, c/d	—	1.50	—
	Shear Buckling Coeff, k_v	IS 800 Cl. 8.4.2.2	7.13	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3} \tau_{cr})}$	2.27	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	39.26 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	931.15 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.65	PASS
G2	Applied Shear, V_u	Governing LC (ULS)	603.36 kN	—
	Shear Area, A_v	$d_w \times t_w$	26,088 mm ²	—
	Panel Aspect Ratio, c/d	—	1.50	—
	Shear Buckling Coeff, k_v	IS 800 Cl. 8.4.2.2	7.13	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3} \tau_{cr})}$	2.27	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	39.26 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	931.15 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.65	PASS
G3	Applied Shear, V_u	Governing LC (ULS)	603.36 kN	—
	Shear Area, A_v	$d_w \times t_w$	26,088 mm ²	—
	Panel Aspect Ratio, c/d	—	1.50	—

Shear Capacity Check (continued)

Girder	Parameter	Reference	Value	Status
	Shear Buckling Coeff, k_v	IS 800 Cl. 8.4.2.2	7.13	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\tau_{cr})}$	2.27	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	39.26 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	931.15 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.65	PASS
G4	Applied Shear, V_u	Governing LC (ULS)	603.36 kN	—
	Shear Area, A_v	$d_w \times t_w$	26,088 mm ²	—
	Panel Aspect Ratio, c/d	—	1.50	—
	Shear Buckling Coeff, k_v	IS 800 Cl. 8.4.2.2	7.13	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\tau_{cr})}$	2.27	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	39.26 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	931.15 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.65	PASS
G5	Applied Shear, V_u	Governing LC (ULS)	603.36 kN	—
	Shear Area, A_v	$d_w \times t_w$	26,088 mm ²	—
	Panel Aspect Ratio, c/d	—	1.50	—
	Shear Buckling Coeff, k_v	IS 800 Cl. 8.4.2.2	7.13	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\tau_{cr})}$	2.27	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	39.26 MPa	—

Shear Capacity Check (continued)

Girder	Parameter	Reference	Value	Status
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	931.15 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.65	PASS
G6	Applied Shear, V_u	Governing LC (ULS)	603.36 kN	—
	Shear Area, A_v	$d_w \times t_w$	26,088 mm ²	—
	Panel Aspect Ratio, c/d	—	1.50	—
	Shear Buckling Coeff, k_v	IS 800 Cl. 8.4.2.2	7.13	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3} \tau_{cr})}$	2.27	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	39.26 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	931.15 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.65	PASS
G7	Applied Shear, V_u	Governing LC (ULS)	603.36 kN	—
	Shear Area, A_v	$d_w \times t_w$	26,088 mm ²	—
	Panel Aspect Ratio, c/d	—	1.50	—
	Shear Buckling Coeff, k_v	IS 800 Cl. 8.4.2.2	7.13	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3} \tau_{cr})}$	2.27	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	39.26 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	931.15 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.65	PASS
G8	Applied Shear, V_u	Governing LC (ULS)	603.36 kN	—

Shear Capacity Check (continued)

Girder	Parameter	Reference	Value	Status
	Shear Area, A_v	$d_w \times t_w$	26,088 mm ²	—
	Panel Aspect Ratio, c/d	—	1.50	—
	Shear Buckling Coeff, k_v	IS 800 Cl. 8.4.2.2	7.13	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\tau_{cr})}$	2.27	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	39.26 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	931.15 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.65	PASS
G9	Applied Shear, V_u	Governing LC (ULS)	603.36 kN	—
	Shear Area, A_v	$d_w \times t_w$	26,088 mm ²	—
	Panel Aspect Ratio, c/d	—	1.50	—
	Shear Buckling Coeff, k_v	IS 800 Cl. 8.4.2.2	7.13	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\tau_{cr})}$	2.27	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	39.26 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	931.15 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.65	PASS
G10	Applied Shear, V_u	Governing LC (ULS)	603.36 kN	—
	Shear Area, A_v	$d_w \times t_w$	26,088 mm ²	—
	Panel Aspect Ratio, c/d	—	1.50	—
	Shear Buckling Coeff, k_v	IS 800 Cl. 8.4.2.2	7.13	—

Shear Capacity Check (continued)

Girder	Parameter	Reference	Value	Status
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\tau_{cr})}$	2.27	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	39.26 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	931.15 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.65	PASS

Note: IS 800 Cl. 8.4, IRC 22 Cl. 603.3.3.2

Table 5.5: Interaction Checks (M-V and M-N)

Girder	Check	Condition	Value	Status
G1	High Shear Condition?	$V_u > 0.6 V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	23,734.4 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.23	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	$0.00 + 0.22 = 0.227$	0.227	PASS
G2	High Shear Condition?	$V_u > 0.6 V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	23,734.4 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.23	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	$0.00 + 0.22 = 0.227$	0.227	PASS
G3	High Shear Condition?	$V_u > 0.6 V_d$	No	—

Interaction Checks (M-V and M-N) (continued)

Girder	Check	Condition	Value	Status
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	23,734.4 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.23	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	$0.00 + 0.22 = 0.227$	0.227	PASS
G4	High Shear Condition?	$V_u > 0.6 V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	23,734.4 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.23	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	$0.00 + 0.22 = 0.227$	0.227	PASS
G5	High Shear Condition?	$V_u > 0.6 V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	23,734.4 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.23	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	$0.00 + 0.22 = 0.227$	0.227	PASS
G6	High Shear Condition?	$V_u > 0.6 V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	23,734.4 kN-m	—

Interaction Checks (M-V and M-N) (continued)

Girder	Check	Condition	Value	Status
	Interaction Check: $M_u \leq M_{dv}$	—	0.23	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	$0.00 + 0.22 = 0.227$	0.227	PASS
G7	High Shear Condition?	$V_u > 0.6 V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	23,734.4 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.23	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	$0.00 + 0.22 = 0.227$	0.227	PASS
G8	High Shear Condition?	$V_u > 0.6 V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	23,734.4 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.23	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	$0.00 + 0.22 = 0.227$	0.227	PASS
G9	High Shear Condition?	$V_u > 0.6 V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	23,734.4 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.23	PASS

Interaction Checks (M-V and M-N) (continued)

Girder	Check	Condition	Value	Status
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	$0.00 + 0.22 = 0.227$	0.227	PASS
G10	High Shear Condition?	$V_u > 0.6 V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	23,734.4 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.23	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	$0.00 + 0.22 = 0.227$	0.227	PASS

Note: IRC 22 Cl. 603.3.3.3

Table 5.6: Lateral Torsional Buckling Check – Construction Stage

Girder	Parameter	Reference	Value	Status
G1	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	8003.59 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	1.48	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.32	—
	LTB Resistance, M_b	$\chi_{LT} M_p/\gamma_{m0}$	5676.22 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.64	PASS
G2	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	8003.59 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	1.48	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.32	—
	LTB Resistance, M_b	$\chi_{LT} M_p/\gamma_{m0}$	5676.22 kN-m	—

Lateral Torsional Buckling Check – Construction Stage (continued)

Girder	Parameter	Reference	Value	Status
	$M_u \leq M_b$	M_u/M_b	0.64	PASS
G3	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	8003.59 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	1.48	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.32	—
	LTB Resistance, M_b	$\chi_{LT} M_p/\gamma_{m0}$	5676.22 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.64	PASS
G4	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	8003.59 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	1.48	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.32	—
	LTB Resistance, M_b	$\chi_{LT} M_p/\gamma_{m0}$	5676.22 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.64	PASS
G5	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	8003.59 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	1.48	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.32	—
	LTB Resistance, M_b	$\chi_{LT} M_p/\gamma_{m0}$	5676.22 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.64	PASS
G6	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	8003.59 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	1.48	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.32	—
	LTB Resistance, M_b	$\chi_{LT} M_p/\gamma_{m0}$	5676.22 kN-m	—

Lateral Torsional Buckling Check – Construction Stage (continued)

Girder	Parameter	Reference	Value	Status
	$M_u \leq M_b$	M_u/M_b	0.64	PASS
G7	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	8003.59 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	1.48	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.32	—
	LTB Resistance, M_b	$\chi_{LT} M_p/\gamma_{m0}$	5676.22 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.64	PASS
G8	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	8003.59 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	1.48	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.32	—
	LTB Resistance, M_b	$\chi_{LT} M_p/\gamma_{m0}$	5676.22 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.64	PASS
G9	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	8003.59 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	1.48	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.32	—
	LTB Resistance, M_b	$\chi_{LT} M_p/\gamma_{m0}$	5676.22 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.64	PASS
G10	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	8003.59 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	1.48	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	0.32	—
	LTB Resistance, M_b	$\chi_{LT} M_p/\gamma_{m0}$	5676.22 kN-m	—

Lateral Torsional Buckling Check – Construction Stage (continued)

Girder	Parameter	Reference	Value	Status
	$M_u \leq M_b$	M_u/M_b	0.64	PASS

Note: IRC 22 Cl. 603.3.3.1, IS 800 Cl. 8.2.2

Table 5.7: Stiffener Design Summary

Girder	Parameter	Value
G1	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	27.8
	Intermediate Stiffener Spacing (mm)	—
	End Panel Stiffener Thickness (mm)	1.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G2	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	27.8
	Intermediate Stiffener Spacing (mm)	—
	End Panel Stiffener Thickness (mm)	1.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G3	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	27.8
	Intermediate Stiffener Spacing (mm)	—
	End Panel Stiffener Thickness (mm)	1.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G4	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	27.8

Stiffener Design Summary (continued)

Girder	Parameter	Value
	Intermediate Stiffener Spacing (mm)	—
	End Panel Stiffener Thickness (mm)	1.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G5	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	27.8
	Intermediate Stiffener Spacing (mm)	—
	End Panel Stiffener Thickness (mm)	1.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G6	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	27.8
	Intermediate Stiffener Spacing (mm)	—
	End Panel Stiffener Thickness (mm)	1.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G7	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	27.8
	Intermediate Stiffener Spacing (mm)	—
	End Panel Stiffener Thickness (mm)	1.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G8	Shear Buckling Design Method	post_critical

Stiffener Design Summary (continued)

Girder	Parameter	Value
	Intermediate Stiffener Thickness (mm)	27.8
	Intermediate Stiffener Spacing (mm)	—
	End Panel Stiffener Thickness (mm)	1.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G9	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	27.8
	Intermediate Stiffener Spacing (mm)	—
	End Panel Stiffener Thickness (mm)	1.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G10	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	27.8
	Intermediate Stiffener Spacing (mm)	—
	End Panel Stiffener Thickness (mm)	1.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None

Table 5.8: End Panel Stiffener Checks

Girder	Check	Required	Provided	Status
G1	Web Buckling Resistance	732.15 kN	206,065.23 kN	PASS
	Local Crushing Resistance	732.15 kN	892.51 kN	PASS
	Bearing Capacity	732.15 kN	4136.36 kN	PASS

End Panel Stiffener Checks (continued)

Girder	Check	Required	Provided	Status
	Column Buckling Resistance	732.15 kN	3900.91 kN	PASS
G2	Web Buckling Resistance	732.15 kN	206,065.23 kN	PASS
	Local Crushing Resistance	732.15 kN	892.51 kN	PASS
	Bearing Capacity	732.15 kN	4136.36 kN	PASS
	Column Buckling Resistance	732.15 kN	3900.91 kN	PASS
G3	Web Buckling Resistance	732.15 kN	206,065.23 kN	PASS
	Local Crushing Resistance	732.15 kN	892.51 kN	PASS
	Bearing Capacity	732.15 kN	4136.36 kN	PASS
	Column Buckling Resistance	732.15 kN	3900.91 kN	PASS
G4	Web Buckling Resistance	732.15 kN	206,065.23 kN	PASS
	Local Crushing Resistance	732.15 kN	892.51 kN	PASS
	Bearing Capacity	732.15 kN	4136.36 kN	PASS
	Column Buckling Resistance	732.15 kN	3900.91 kN	PASS
G5	Web Buckling Resistance	732.15 kN	206,065.23 kN	PASS
	Local Crushing Resistance	732.15 kN	892.51 kN	PASS
	Bearing Capacity	732.15 kN	4136.36 kN	PASS
	Column Buckling Resistance	732.15 kN	3900.91 kN	PASS
G6	Web Buckling Resistance	732.15 kN	206,065.23 kN	PASS
	Local Crushing Resistance	732.15 kN	892.51 kN	PASS
	Bearing Capacity	732.15 kN	4136.36 kN	PASS

End Panel Stiffener Checks (continued)

Girder	Check	Required	Provided	Status
	Column Buckling Resistance	732.15 kN	3900.91 kN	PASS
G7	Web Buckling Resistance	732.15 kN	206,065.23 kN	PASS
	Local Crushing Resistance	732.15 kN	892.51 kN	PASS
	Bearing Capacity	732.15 kN	4136.36 kN	PASS
	Column Buckling Resistance	732.15 kN	3900.91 kN	PASS
G8	Web Buckling Resistance	732.15 kN	206,065.23 kN	PASS
	Local Crushing Resistance	732.15 kN	892.51 kN	PASS
	Bearing Capacity	732.15 kN	4136.36 kN	PASS
	Column Buckling Resistance	732.15 kN	3900.91 kN	PASS
G9	Web Buckling Resistance	732.15 kN	206,065.23 kN	PASS
	Local Crushing Resistance	732.15 kN	892.51 kN	PASS
	Bearing Capacity	732.15 kN	4136.36 kN	PASS
	Column Buckling Resistance	732.15 kN	3900.91 kN	PASS
G10	Web Buckling Resistance	732.15 kN	206,065.23 kN	PASS
	Local Crushing Resistance	732.15 kN	892.51 kN	PASS
	Bearing Capacity	732.15 kN	4136.36 kN	PASS
	Column Buckling Resistance	732.15 kN	3900.91 kN	PASS

Note: IS 800 Cl. 8.4.2.2

Table 5.9: Serviceability – Deflection Checks

Girder	Check	Allowable	Actual	Status
G1	Live Load Deflection, δ_{LL} (mm)	$L/800 = 50$ mm	16.458	PASS
	Total Load Deflection, δ_{total} (mm)	$L/600 = 66.667$ mm	70.915	FAIL
G2	Live Load Deflection, δ_{LL} (mm)	$L/800 = 50$ mm	16.734	PASS
	Total Load Deflection, δ_{total} (mm)	$L/600 = 66.667$ mm	71.461	FAIL
G3	Live Load Deflection, δ_{LL} (mm)	$L/800 = 50$ mm	17.002	PASS
	Total Load Deflection, δ_{total} (mm)	$L/600 = 66.667$ mm	71.923	FAIL
G4	Live Load Deflection, δ_{LL} (mm)	$L/800 = 50$ mm	17.196	PASS
	Total Load Deflection, δ_{total} (mm)	$L/600 = 66.667$ mm	72.258	FAIL
G5	Live Load Deflection, δ_{LL} (mm)	$L/800 = 50$ mm	17.298	PASS
	Total Load Deflection, δ_{total} (mm)	$L/600 = 66.667$ mm	72.433	FAIL
G6	Live Load Deflection, δ_{LL} (mm)	$L/800 = 50$ mm	17.298	PASS
	Total Load Deflection, δ_{total} (mm)	$L/600 = 66.667$ mm	72.432	FAIL

Serviceability – Deflection Checks (continued)

Girder	Check	Allowable	Actual	Status
G7	Live Load Deflection, δ_{LL} (mm)	$L/800 = 50$ mm	17.196	PASS
	Total Load Deflection, δ_{total} (mm)	$L/600 = 66.667$ mm	72.255	FAIL
G8	Live Load Deflection, δ_{LL} (mm)	$L/800 = 50$ mm	17.002	PASS
	Total Load Deflection, δ_{total} (mm)	$L/600 = 66.667$ mm	71.919	FAIL
G9	Live Load Deflection, δ_{LL} (mm)	$L/800 = 50$ mm	16.734	PASS
	Total Load Deflection, δ_{total} (mm)	$L/600 = 66.667$ mm	71.455	FAIL
G10	Live Load Deflection, δ_{LL} (mm)	$L/800 = 50$ mm	16.458	PASS
	Total Load Deflection, δ_{total} (mm)	$L/600 = 66.667$ mm	70.908	FAIL

Note: IRC 22 Cl. 604.3.2

Table 5.10: Serviceability – Maximum Stress Limitation

Girder	Element	Allowable Stress	Actual Stress	Status
G1	Structural Steel ($0.9 f_y$)	315 MPa	89.823 MPa	PASS
G2	Structural Steel ($0.9 f_y$)	315 MPa	89.823 MPa	PASS
G3	Structural Steel ($0.9 f_y$)	315 MPa	89.823 MPa	PASS

Serviceability – Maximum Stress Limitation (continued)

Girder	Element	Allowable Stress	Actual Stress	Status
G4	Structural Steel ($0.9 f_y$)	315 MPa	89.823 MPa	PASS
G5	Structural Steel ($0.9 f_y$)	315 MPa	89.823 MPa	PASS
G6	Structural Steel ($0.9 f_y$)	315 MPa	89.823 MPa	PASS
G7	Structural Steel ($0.9 f_y$)	315 MPa	89.823 MPa	PASS
G8	Structural Steel ($0.9 f_y$)	315 MPa	89.823 MPa	PASS
G9	Structural Steel ($0.9 f_y$)	315 MPa	89.823 MPa	PASS
G10	Structural Steel ($0.9 f_y$)	315 MPa	89.823 MPa	PASS

Table 5.11: Serviceability – Fatigue Assessment

Girder	Stress Range, $\Delta\sigma$ (MPa)	Fatigue Limit, f_{fd} (MPa)	Utilization Ratio	Status
G1	160.07 MPa	89.91 MPa	1.78	FAIL
G2	160.33 MPa	89.91 MPa	1.78	FAIL
G3	161.02 MPa	89.91 MPa	1.79	FAIL
G4	161.7 MPa	89.91 MPa	1.80	FAIL
G5	162.31 MPa	89.91 MPa	1.81	FAIL
G6	162.63 MPa	89.91 MPa	1.81	FAIL
G7	162.63 MPa	89.91 MPa	1.81	FAIL
G8	162.3 MPa	89.91 MPa	1.81	FAIL
G9	161.69 MPa	89.91 MPa	1.80	FAIL
G10	161.01 MPa	89.91 MPa	1.79	FAIL

Note: IRC 22 Cl. 605 — governing of normal and shear fatigue (worst by DCR). Capacity reduction factor μ_r applied where plate thickness > 25 mm.

Table 5.12: Girder Design Summary (DCR / Utilization Ratio)

Girder	Controlling LC / Combination	Controlling Check	Demand	Capacity	UR	Status
G1	SLS.FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	160.07 MPa	89.91 MPa	1.780	FAIL
G2	SLS.FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	160.33 MPa	89.91 MPa	1.783	FAIL
G3	SLS.FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	161.02 MPa	89.91 MPa	1.791	FAIL
G4	SLS.FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	161.7 MPa	89.91 MPa	1.798	FAIL
G5	SLS.FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	162.31 MPa	89.91 MPa	1.805	FAIL
G6	SW	Int.Stiff: Leg $\leq H_{limit}$	325 mm	169.03 mm	1.923	FAIL
G7	SW	Int.Stiff: Leg $\leq H_{limit}$	325 mm	169.03 mm	1.923	FAIL
G8	SW	Int.Stiff: Leg $\leq H_{limit}$	325 mm	169.03 mm	1.923	FAIL
G9	SW	Int.Stiff: Leg $\leq H_{limit}$	325 mm	169.03 mm	1.923	FAIL
G10	SW	Int.Stiff: Leg $\leq H_{limit}$	325 mm	169.03 mm	1.923	FAIL

Note: UR = Demand / Capacity. A value ≤ 1.0 indicates a passing check. The controlling check is the criterion with the highest UR for each girder, with the real load case/combination that drives it.

Table 5.13: Shear Connector Capacity (bridge-level)

Parameter	Formula	Value	Reference
Design Resistance, Q_u	$Q_u = \min(Q_{u,s}, Q_{u,c})$ $Q_{u,s} = \frac{0.8 f_u (\pi d^2 / 4)}{\gamma_v}$ $Q_{u,c} = \frac{0.29 \alpha d^2 \sqrt{f_{ck} E_{cm}}}{\gamma_v}$	95.363 kN	IRC 22 Cl. 606.3.1 (Eq. 6.1)
Fatigue Shear Resistance, Q_r	IRC 22 Table 8 ($\phi d, N_{sc}$)	25 kN	IRC 22 Cl. 606.3.2 (Table 8)

Table 5.14: Shear Connector Spacing

Criterion	Governing Spacing	Actual Spacing Provided	Status
ULS Shear (SL1)	2279.53 mm	400 mm	PASS
Full Composite (SL2)	429.85 mm	400 mm	PASS
SLS Fatigue (SR)	837.04 mm	400 mm	PASS
Max Spacing Limit (IRC 22)	400 mm	400 mm	PASS

Note: IRC 22 Cl. 606.4, 606.9. Governing spacing = $\min(S_{L1}, S_{L2}, S_R)$.

Table 5.15: Transverse Shear and Detailing Checks

Check	Value	Status
Longitudinal Shear per unit length, V_L	83.669 kN/m	—
Transverse Shear Capacity of Slab, V_{Rd}	999.28 kN/m	—
Transverse Shear Check	$V_L / V_{Rd} = 0.08$	PASS
Min. Transverse Reinforcement, $A_{st,min}$	Required 0.418 cm ² /m, Provided 11.57 cm ² /m	PASS
Stud Diameter $\leq 2 t_f$	$d = 20 \text{ mm} \leq 2 t_f = 56 \text{ mm}$	PASS
Stud Edge Distance	Provided 275 mm (req. $\geq 25 \text{ mm}$)	PASS

Note: IRC 22 Cl. 606.6, 606.10.

5.2 Deck Slab Design

The reinforced concrete deck slab is designed per IRC 112:2011 (flexure, shear, crack width) and IRC 22:2014 (composite construction). Wheel loads are distributed using Pigeaud's method. The deck is checked for flexure in the transverse and longitudinal directions, punching shear, one-way (beam) shear, crack width, and reinforcement detailing.

Table 5.16: Deck Slab — Loading and Geometry

Parameter	Value
Effective Span of Deck Slab, l_{eff}	2.465 m (girder spacing, c/c)
Deck Thickness, t_s	250 mm
Clear Cover (IRC 112 Cl. 15.2)	Top 50 mm / Bottom 50 mm
Concrete Grade (IRC 112 Cl. 6.4)	M40 ($f_{ck} = 40$ MPa, $f_{ctm} = 3$ MPa)
Reinforcement Grade (IRC 112 Cl. 6.2)	Fe 500 ($f_y = 500$ MPa)
Dead Load per Unit Area, w_{DL}	6.131 kN/m ² (slab self-weight)
IRC 6 Wheel Load (Class A / 70R)	83.385 kN
Tyre Contact Width (IRC 6 Annex A)	300 mm (transverse)
Impact Factor (IRC 6 Cl. 208.2)	0.100
Governing Live Load Case	Class70R(W)

Table 5.17: Deck Slab — Flexure Check: Interior Panel (Pigeaud's Method)

Location	Parameter	Formula / Reference	Value	Status
At Midspan (Sagging)	Transverse BM (DL), $M_{T,DL}$	$w_{DL} l_{eff}^2 / 10$	3.725 kN-m/m	—
At Midspan (Sagging)	Transverse BM (LL), $M_{T,LL}$	Effective width (IRC 112 B3.1)	25.664 kN-m/m	—
At Midspan (Sagging)	Total Design BM, $M_{u,sag}$	1.35 DL + 1.50 LL	47.375 kN-m/m	—
At Midspan (Sagging)	Effective depth, d	$t_s - c_{nom} - \phi/2$	194 mm	—
At Midspan (Sagging)	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	53.7 kN-m/m	PASS
At Support (Hogging)	Total Design BM, $M_{u,hog}$	1.35 DL + 1.50 LL (at support)	35.531 kN-m/m	—

Deck Slab — Flexure Check: Interior Panel (Pigeaud's Method) (continued)

Location	Parameter	Formula / Reference	Value	Status
At Support (Hogging)	Required Top Steel, $A_{st,top}$	$M_u / (0.87 f_y d)$	487.5 mm ² /m	—
At Support (Hogging)	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	40.162 kN-m/m	PASS

Note: IRC 112 Cl. 12.2. Distribution (longitudinal) reinforcement designed for 20% of main steel moment (IRC 21 Cl. 305.18).

Table 5.18: Deck Slab — Cantilever Overhang Flexure Check

Parameter	Formula	Value	Status
Overhang Length, l_{oh}	—	1.232 m	—
Crash Barrier Load Moment	IRC 6 Cl. 206.4	7.5 kN-m/m	—
Dead Load Moment	$w_{DL} l_{oh}^2 / 2 + \text{railing}$	6.47 kN-m/m	—
Live Load Moment (eccentric wheel)	Wheel load $\times$ arm	53.128 kN-m/m	—
Total Hogging Moment, $M_{u,oh}$	1.35 DL + 1.50 (LL + CB)	107.646 kN-m/m	—
Moment Capacity (top steel), $M_{Rd,oh}$	IRC 112 Cl. 12.2	114.707 kN-m/m	PASS

Note: IRC 6 Cl. 206.4 crash barrier loads applied at kerb face; IRC 112 Cl. 12.2 flexure.

Table 5.19: Deck Slab — Punching Shear Check (IRC 112 Cl. 10.4.6)

Parameter	Formula / Reference	Value	Status
Design Wheel Load (ULS), V_{Ed}	$\gamma_Q (1 + IF) P_w$	137.585 kN	—
Tyre Contact Area	$a \times b$ (IRC 6 Annex A)	300 $\times$ 150 mm	—
Loaded Area at mid-depth, b_0	$c_1 \times c_2$ (incl. WC dispersion)	400 $\times$ 250 mm	—
Control Perimeter, u_1	$2(c_1 + c_2) + 4\pi d$	3737.876 mm	—
Punching Shear Stress, v_{Ed}	$V_{Ed} / (u_1 d)$	0.1897 MPa	—
Punching Resistance, $v_{Rd,c}$	IRC 112 Eq. 10.1	0.5545 MPa	—

Deck Slab — Punching Shear Check (IRC 112 Cl. 10.4.6) (continued)

Parameter	Formula / Reference	Value	Status
Punching Shear Check	$v_{Ed} \leq v_{Rd,c}$	0.34	PASS

Note: Punching shear reinforcement not typically required for deck slabs with $d \geq 200$ mm and adequate longitudinal reinforcement.

Table 5.20: Crack Width Check (Deck Slab)

Parameter	Value / Reference
Min. Reinforcement for Crack Control, $A_{s,min}$	302.64 mm ² /m [IRC 112 Cl. 16.5.1]
Provided Reinforcement (bottom)	$\phi 12$ @ 170 mm c/c (665.278 mm ² /m)
Max. Permissible Crack Width	0.3 mm
Calculated Crack Width, w_k (governing)	0.2218 mm
Crack Width Check	PASS

Table 5.21: One-Way (Beam) Shear Check (Deck Slab)

Parameter	Formula / Reference	Value	Status
Design Shear per unit width, V_{Ed}	$\gamma_{DL}V_{DL} + \gamma_{LL}(1+IF)V_{LL}$	78.917 kN/m	—
Effective depth, d	$t_s - c_{nom} - \phi/2$	194 mm	—
Size factor, k	$1 + \sqrt{200/d} \leq 2.0$	2.000	—
Longitudinal reinforcement ratio, ρ_l	$A_{sl}/(b_w d) \leq 0.02$	0.0034	—
Shear resistance (no stirrups), $V_{Rd,c}$	$v_{Rd,c} b_w d$ (Cl. 10.3.2)	107.582 kN/m	—
One-Way Shear Check	$V_{Ed} \leq V_{Rd,c}$	0.73	PASS

Note: IRC 112 Cl. 10.3.2. Shear reinforcement not provided in deck slabs; capacity relies on concrete and main reinforcement.

Table 5.22: Reinforcement Detailing Summary (Deck Slab)

Parameter	Required / Limit	Provided	Status
Main Reinforcement — Bottom (Transverse)			

Reinforcement Detailing Summary (Deck Slab) (continued)

Parameter	Required / Limit	Provided	Status
Required Area, $A_{st,req}$ (mm ² /m)	615.725 mm ² /m	665.278 mm ² /m	PASS
Bar Diameter × Spacing	$\phi \geq 10$ mm (IRC 112)	$\phi 12 @ 170$ mm c/c	—
Min. Reinforcement $A_{s,min}$ (IRC 112 Cl. 16.3.1)	302.64 mm ² /m	665.278 mm ² /m	PASS
Max. Bar Spacing (IRC 112 Cl. 16.3.2)	300 mm	170 mm	PASS
Distribution Reinforcement — Longitudinal			
Required Area, $A_{st,dist}$ (mm ² /m)	$\geq 20\%$ of main steel	376.991 mm ² /m	PASS
Top Reinforcement (Support / Cantilever Overhang)			
Required Area, $A_{st,top}$ (mm ² /m)	487.5 mm ² /m	491.728 mm ² /m	PASS
Cover and Detailing			
Clear Cover (IRC 112 Cl. 15.2)	≥ 50 mm (Table 14.2)	Top 50 mm / Bottom 50 mm	PASS

Note: IRC 112 Cl. 16.3, IS 456 Cl. 26.5. All reinforcement provisions satisfy strength and detailing requirements.

5.3 Cross Bracing Design

Cross bracing between adjacent plate girders provides lateral stability during construction, resists transverse loads (wind, seismic, braking) in service, and prevents lateral torsional buckling of the girders. Members are designed per IS 800:2007 Cl. 7 (compression) and Cl. 6 (tension). Forces are derived from the grillage model under the governing load combination (DL + LL + WL).

Table 5.23: Cross Bracing — Connection and Section Properties

Panel	Member	Connection	Section	A_g (mm ²)	r_{min} (mm)
G1-G2	Diagonal	Bolted	40 x 40 x 3	237	8
G1-G2	Chord	Bolted	40 x 40 x 3	237	8
G2-G3	Diagonal	Bolted	40 x 40 x 3	237	8
G2-G3	Chord	Bolted	40 x 40 x 3	237	8
G3-G4	Diagonal	Bolted	40 x 40 x 3	237	8
G3-G4	Chord	Bolted	40 x 40 x 3	237	8
G4-G5	Diagonal	Bolted	40 x 40 x 3	237	8
G4-G5	Chord	Bolted	40 x 40 x 3	237	8
G5-G6	Diagonal	Bolted	40 x 40 x 3	237	8
G5-G6	Chord	Bolted	40 x 40 x 3	237	8
G6-G7	Diagonal	Bolted	40 x 40 x 3	237	8
G6-G7	Chord	Bolted	40 x 40 x 3	237	8
G7-G8	Diagonal	Bolted	40 x 40 x 3	237	8
G7-G8	Chord	Bolted	40 x 40 x 3	237	8
G8-G9	Diagonal	Bolted	40 x 40 x 3	237	8
G8-G9	Chord	Bolted	40 x 40 x 3	237	8
G9-G10	Diagonal	Bolted	40 x 40 x 3	237	8

G9-G10	Chord	Bolted	40 x 40 x 3	237	8
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Note: A_g = gross cross-sectional area; r_{min} = minimum radius of gyration.

Table 5.24: Cross Bracing — Slenderness Ratio Check

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit	Status
G1-G2	Diagonal	C	3109.5	131.3	250	PASS
G1-G2	Top chord	C	2465	131.3	250	PASS
G1-G2	Bottom chord	T	2465	200.4	400	PASS
G2-G3	Diagonal	C	3109.5	146.5	250	PASS
G2-G3	Top chord	C	2465	130.3	250	PASS
G2-G3	Bottom chord	T	2465	200.4	400	PASS
G3-G4	Diagonal	C	3109.5	164.3	250	PASS
G3-G4	Top chord	C	2465	130.3	250	PASS
G3-G4	Bottom chord	T	2465	200.4	400	PASS
G4-G5	Diagonal	C	3109.5	164.3	250	PASS
G4-G5	Top chord	C	2465	130.3	250	PASS
G4-G5	Bottom chord	T	2465	200.4	400	PASS
G5-G6	Diagonal	C	3109.5	164.3	250	PASS
G5-G6	Top chord	C	2465	130.3	250	PASS
G5-G6	Bottom chord	T	2465	200.4	400	PASS
G6-G7	Diagonal	C	3109.5	164.3	250	PASS
G6-G7	Top chord	C	2465	130.3	250	PASS

Cross Bracing — Slenderness Ratio Check (continued)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit	Status
G6-G7	Bottom chord	T	2465	200.4	400	PASS
G7-G8	Diagonal	C	3109.5	164.3	250	PASS
G7-G8	Top chord	C	2465	130.3	250	PASS
G7-G8	Bottom chord	T	2465	200.4	400	PASS
G8-G9	Diagonal	C	3109.5	164.3	250	PASS
G8-G9	Top chord	C	2465	130.3	250	PASS
G8-G9	Bottom chord	T	2465	200.4	400	PASS
G9-G10	Diagonal	C	3109.5	164.3	250	PASS
G9-G10	Top chord	C	2465	130.3	250	PASS
G9-G10	Bottom chord	T	2465	200.4	400	PASS

Note: Limit = 250 for compression members, 400 for tension members. $K = 1.0$ for members with both ends pinned.

Table 5.25: Cross Bracing Design — Capacity Summary

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G1-G2	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	22.581	45	0.50	PASS
G1-G2	Diagonal (Compression)	50 x 50 x 4	1.5 EQ (b)	27.534	28.69	0.95	PASS
G2-G3	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	18.017	45	0.40	PASS

Cross Bracing Design — Capacity Summary (continued)

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G2-G3	Chord (Compression)	40 x 40 x 3	1.5 EQ (b)	14.486	17.55	0.89	PASS
G3-G4	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	17.669	45	0.39	PASS
G3-G4	Chord (Compression)	40 x 40 x 3	1.5 EQ (b)	8.975	17.55	0.83	PASS
G4-G5	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	17.682	45	0.39	PASS
G4-G5	Diagonal (Compression)	40 x 40 x 3	1.5 EQ (b)	5.596	12.07	0.52	PASS
G5-G6	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	17.811	45	0.40	PASS
G5-G6	Diagonal (Compression)	40 x 40 x 3	1.5 EQ (a)	1.709	12.07	0.16	PASS
G6-G7	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	18.832	45	0.42	PASS
G6-G7	Diagonal (Compression)	40 x 40 x 3	1.5 EQ (b)	1.793	12.07	0.17	PASS
G7-G8	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	20.352	45	0.45	PASS

Cross Bracing Design — Capacity Summary (continued)

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G7-G8	Diagonal (Compression)	40 x 40 x 3	1.5 EQ (b)	4.009	12.07	0.37	PASS
G8-G9	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	22.414	45	0.50	PASS
G8-G9	Diagonal (Compression)	40 x 40 x 3	1.5 EQ (b)	5.815	12.07	0.54	PASS
G9-G10	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (b)	29.545	45	0.66	PASS
G9-G10	Diagonal (Compression)	40 x 40 x 3	1.5 EQ (b)	5.734	12.07	0.53	PASS

Note: Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension). OsdagBridge cross-bracing module used.

5.4 End Diaphragm Design

End diaphragms at the supports transfer transverse loads to the bearings, restrain the bottom flanges against lateral displacement, and maintain the girder cross-section geometry during construction and in service. They are designed per IS 800:2007 and IRC 24:2010 Cl. 507.

Table 5.26: End Diaphragm — Connection and Section Properties

Panel	Member	Connection	Section	A_g (mm ²)	r_{min} (mm)
G1-G2	Diagonal	Bolted	40 x 40 x 3	237	8
G1-G2	Chord	Bolted	40 x 40 x 3	237	8
G2-G3	Diagonal	Bolted	40 x 40 x 3	237	8
G2-G3	Chord	Bolted	40 x 40 x 3	237	8
G3-G4	Diagonal	Bolted	40 x 40 x 3	237	8
G3-G4	Chord	Bolted	40 x 40 x 3	237	8
G4-G5	Diagonal	Bolted	40 x 40 x 3	237	8
G4-G5	Chord	Bolted	40 x 40 x 3	237	8
G5-G6	Diagonal	Bolted	40 x 40 x 3	237	8
G5-G6	Chord	Bolted	40 x 40 x 3	237	8
G6-G7	Diagonal	Bolted	40 x 40 x 3	237	8
G6-G7	Chord	Bolted	40 x 40 x 3	237	8
G7-G8	Diagonal	Bolted	40 x 40 x 3	237	8
G7-G8	Chord	Bolted	40 x 40 x 3	237	8
G8-G9	Diagonal	Bolted	40 x 40 x 3	237	8
G8-G9	Chord	Bolted	40 x 40 x 3	237	8
G9-G10	Diagonal	Bolted	40 x 40 x 3	237	8
G9-G10	Chord	Bolted	40 x 40 x 3	237	8

Note: A_g = gross cross-sectional area; r_{min} = minimum radius of gyration.

Table 5.27: End Diaphragm — Slenderness Ratio Check (IS 800 Cl. 3.8)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit	Status
G1-G2	Diagonal	C	3109.5	131.3	250	PASS
G1-G2	Top chord	C	2465	131.3	250	PASS
G1-G2	Bottom chord	T	2465	200.4	400	PASS
G2-G3	Diagonal	C	3109.5	146.5	250	PASS
G2-G3	Top chord	C	2465	130.3	250	PASS
G2-G3	Bottom chord	T	2465	200.4	400	PASS
G3-G4	Diagonal	C	3109.5	164.3	250	PASS
G3-G4	Top chord	C	2465	130.3	250	PASS
G3-G4	Bottom chord	T	2465	200.4	400	PASS
G4-G5	Diagonal	C	3109.5	164.3	250	PASS
G4-G5	Top chord	C	2465	130.3	250	PASS
G4-G5	Bottom chord	T	2465	200.4	400	PASS
G5-G6	Diagonal	C	3109.5	164.3	250	PASS
G5-G6	Top chord	C	2465	130.3	250	PASS
G5-G6	Bottom chord	T	2465	200.4	400	PASS
G6-G7	Diagonal	C	3109.5	164.3	250	PASS
G6-G7	Top chord	C	2465	130.3	250	PASS
G6-G7	Bottom chord	T	2465	200.4	400	PASS
G7-G8	Diagonal	C	3109.5	164.3	250	PASS

End Diaphragm — Slenderness Ratio Check (IS 800 Cl. 3.8) (continued)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit	Status
G7-G8	Top chord	C	2465	130.3	250	PASS
G7-G8	Bottom chord	T	2465	200.4	400	PASS
G8-G9	Diagonal	C	3109.5	164.3	250	PASS
G8-G9	Top chord	C	2465	130.3	250	PASS
G8-G9	Bottom chord	T	2465	200.4	400	PASS
G9-G10	Diagonal	C	3109.5	164.3	250	PASS
G9-G10	Top chord	C	2465	130.3	250	PASS
G9-G10	Bottom chord	T	2465	200.4	400	PASS

Note: Limit = 250 for compression members, 400 for tension members. $K = 1.0$ for members with both ends pinned.

Table 5.28: End Diaphragm Design — Capacity Summary

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G1-G2	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	22.581	45	0.50	PASS
G1-G2	Diagonal (Compression)	50 x 50 x 4	1.5 EQ (b)	27.534	28.69	0.95	PASS
G2-G3	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	18.017	45	0.40	PASS
G2-G3	Chord (Compression)	40 x 40 x 3	1.5 EQ (b)	14.486	17.55	0.89	PASS

End Diaphragm Design — Capacity Summary (continued)

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G3-G4	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	17.669	45	0.39	PASS
G3-G4	Chord (Compression)	40 x 40 x 3	1.5 EQ (b)	8.975	17.55	0.83	PASS
G4-G5	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	17.682	45	0.39	PASS
G4-G5	Diagonal (Compression)	40 x 40 x 3	1.5 EQ (b)	5.596	12.07	0.52	PASS
G5-G6	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	17.811	45	0.40	PASS
G5-G6	Diagonal (Compression)	40 x 40 x 3	1.5 EQ (a)	1.709	12.07	0.16	PASS
G6-G7	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	18.832	45	0.42	PASS
G6-G7	Diagonal (Compression)	40 x 40 x 3	1.5 EQ (b)	1.793	12.07	0.17	PASS
G7-G8	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	20.352	45	0.45	PASS
G7-G8	Diagonal (Compression)	40 x 40 x 3	1.5 EQ (b)	4.009	12.07	0.37	PASS

End Diaphragm Design — Capacity Summary (continued)

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G8-G9	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (a)	22.414	45	0.50	PASS
G8-G9	Diagonal (Compression)	40 x 40 x 3	1.5 EQ (b)	5.815	12.07	0.54	PASS
G9-G10	Diagonal (Tension)	40 x 40 x 3	1.5 EQ (b)	29.545	45	0.66	PASS
G9-G10	Diagonal (Compression)	40 x 40 x 3	1.5 EQ (b)	5.734	12.07	0.53	PASS

Note: Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension). OsdagBridge cross-bracing module used.

5.5 Overall Design Check Summary

Table 5.29: Overall Design Check Summary

Check / Member	Governing LC	Demand	Capacity	UR	Status
Girder — Moment	ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	5272.71 kNm	23,734.4 kNm	0.22	PASS
Girder — Shear	ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	732.15 kN	1113.56 kN	0.66	PASS
Girder — LTB (constr.)	1.0 DL	3610.52 kNm	5676.22 kNm	0.64	PASS
Girder — Deflection	1.0 DL + 1.0 LL	36.28 mm	66.67 mm	0.54	PASS
Girder — Stress	SLS_RARE_1: 1.0DL + 1.2DW + 1.0LL + 0.6WL	92.51 MPa	315 MPa	0.29	PASS
Girder — Fatigue	—	162.63 MPa	89.91 MPa	1.81	FAIL
Crack Width (slab)	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	0.2218 mm	0.3 mm	0.74	PASS
Deck — Flexure (sagging)	Basic ULS: 1.35DL + 1.5LL	47.375 kN-m/m	53.7 kN-m/m	0.88	PASS
Deck — Flexure (hogging)	Basic ULS: 1.35DL + 1.5LL	35.531 kN-m/m	40.162 kN-m/m	0.88	PASS
Deck — Cantilever Overhang	Basic ULS: 1.35DL + 1.5LL	107.646 kN-m/m	114.707 kN-m/m	0.94	PASS
Deck — Punching Shear	Basic ULS: 1.35DL + 1.5LL	0.1897 MPa	0.5545 MPa	0.34	PASS

Overall Design Check Summary (continued)

Check / Member	Governing LC	Demand	Capacity	UR	Status
Deck — One-Way Shear	Basic ULS: 1.35DL + 1.5LL	78.917 kN/m	107.582 kN/m	0.73	PASS
Cross Bracing — Compression	1.5 EQ (b)	27.534 kN	28.69 kN	0.95	PASS
Cross Bracing — Tension	1.5 EQ (b)	29.545 kN	45 kN	0.66	PASS
Cross Bracing — Slenderness	—	—	—	1.01	FAIL
End Diaphragm — Compression	1.5 EQ (b)	27.534 kN	28.69 kN	0.95	PASS
End Diaphragm — Tension	1.5 EQ (b)	29.545 kN	45 kN	0.66	PASS

Note: UR = Demand / Capacity. All values ≤ 1.0 indicate passing checks. The governing check for each component is highlighted in the individual design check sections above.

The chart below illustrates the governing utilization ratios for all primary superstructure components. Ratios exceeding 1.0 indicate demand exceeding capacity under the governing limit state load combinations.

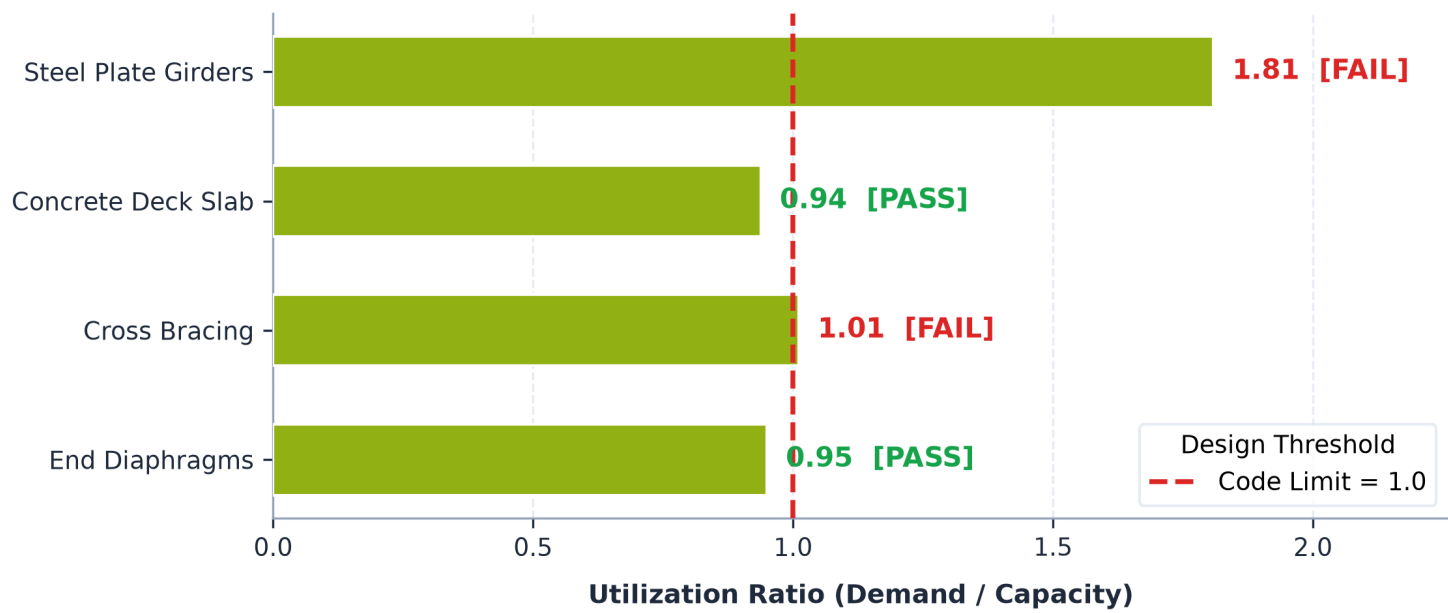


Figure 5.1: Overall Utilization Ratio by Component

6 Drawings and Visualizations

This section presents CAD-generated views of the designed bridge and its components. All views are generated automatically by OsdagBridge using pythonOCC.

6.1 Bridge Configuration and Layout



Figure 6.1: Overall 3D Bridge Superstructure



Figure 6.2: Typical Cross Section



Figure 6.3: Top View

6.2 Plate Girder — Detailed Views



Figure 6.4: 3D View of Plate Girders



Figure 6.5: Cross Section of Plate Girder



Figure 6.6: Side View of Girder

6.3 Cross Bracing Detail



Figure 6.7: Cross Bracing Layout

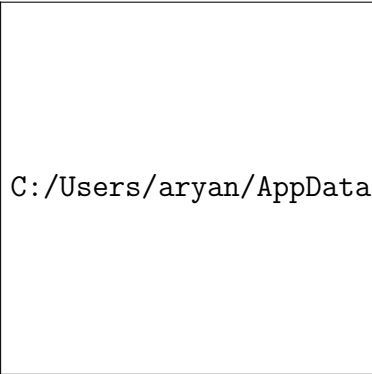


Figure 6.8: Cross Bracing – Bracing Section

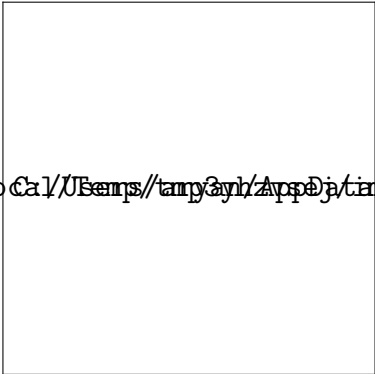


Figure 6.9: Cross Bracing – Top Chord Section

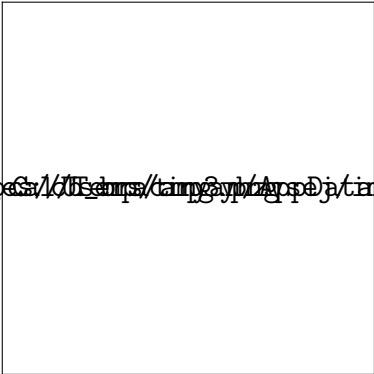


Figure 6.10: Cross Bracing – Bottom Chord Section

6.4 End Diaphragm Detail



Figure 6.11: End Diaphragm Layout

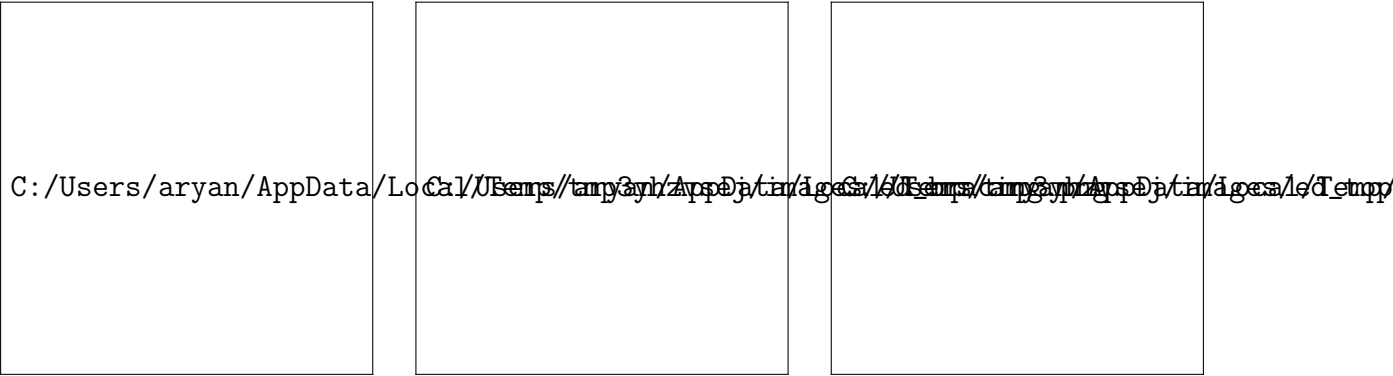


Figure 6.12: End Diaphragm – Bracing Section

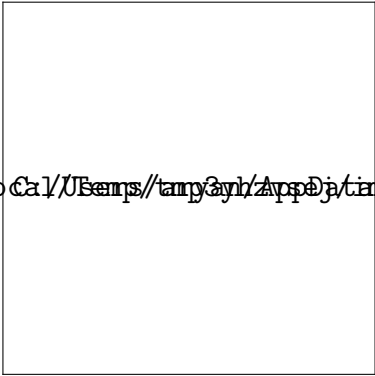


Figure 6.13: End Diaphragm – Top Chord Section

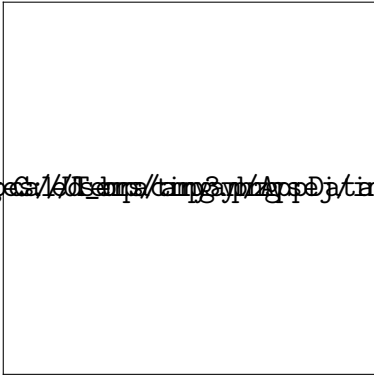


Figure 6.14: End Diaphragm – Bottom Chord Section

7 Material Take-off & Quantity Summary

Table 7.1: Bill of Materials (Steel, Concrete, and Reinforcement Quantities)

S.N.	Item Description	Volume	Quantity	Total Volume	Weight (MT)	Total Weight (MT)
1	Structural Steel (IS 2062) for Girders	$0.06104 \text{ m}^2 \times 40.00 \text{ m} = 2.44173 \text{ m}^3$	10	24.42	19.17	191.68
2(a)	Cross Bracing - Top Chord	$0.00024 \text{ m}^2 \times 2.46 \text{ m} = 0.00058 \text{ m}^3$	9	0.01	0.0046	0.04
2(b)	Cross Bracing - Bottom Chord	$0.00024 \text{ m}^2 \times 2.46 \text{ m} = 0.00058 \text{ m}^3$	9	0.01	0.0046	0.04
2(c)	Cross Bracing - Diagonal Chord	$0.00024 \text{ m}^2 \times 3.11 \text{ m} = 0.00074 \text{ m}^3$	18	0.01	0.0058	0.10
2(d)	End Diaphragm	—	—	—	—	—
3	Concrete (M40) for Deck Slab	$24.65 \text{ m} \times 0.25 \text{ m} \times 40.00 \text{ m} = 246.50000 \text{ m}^3$	1	246.50	616.25	616.25
4	Reinforcement Steel (Fe 500)	$0.09420 \text{ m}^2 \times 40.00 \text{ m} = 3.76815 \text{ m}^3$	1	3.77	29.58	29.58
5	Shear Stud Connectors	$0.00031 \text{ m}^2 \times 0.10 \text{ m} = 0.00003 \text{ m}^3$	2020	0.06	0.000247	0.498
6	Crash Barrier	$0.00000 \text{ m}^2 \times 40.00 \text{ m} = 0.00001 \text{ m}^3$	2	0.00	0.00	0.00

The charts below summarize the distribution of structural steel components and key construction material quantities required for the bridge superstructure.

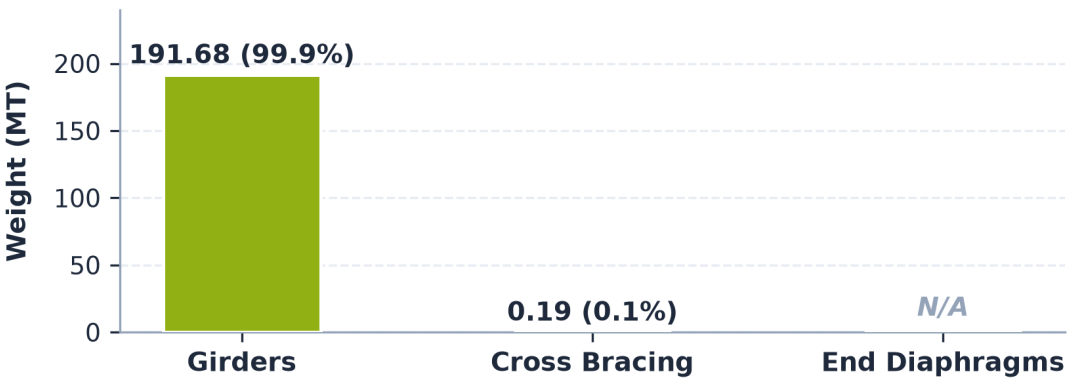


Figure 7.1: Structural Steel Quantities



Figure 7.2: Concrete Volume



Figure 7.3: Reinforcement Steel Quantity

8 Standards & Assumptions

This section provides references to standards used to calibrate the OsdagBridge design modules, and notes the limitations of the current software version.

8.1 Design Standards

The following Indian Road Congress (IRC) codes and Indian Standards (IS) form the basis of all design calculations in this software.

Table 8.1: IRC Codes

Code	Year	Title / Scope
IRC 5	2015	General Features of Design - carriageway widths, kerb, footpath dimensions
IRC 6	2017	Loads and Load Combinations - dead load, live load, impact, wind, temperature, etc.
IRC 22	2015	Composite Construction (LS) - Composite section properties, ULS/SLS design, shear connectors
IRC 24	2010	Steel Road Bridges (LS) - Stiffener design, skew angle limits, diaphragm requirements
IRC 112	2020	Concrete Road Bridges - Deck slab flexure, shear, crack width, reinforcement
IRC SP 114	2018	Seismic Design of Road Bridges

Table 8.2: IS Codes

Code	Year	Scope
IS 800	2007	Steel construction - tension, compression, bending, shear, LTB, stiffeners, combined checks
IS 456	2000	Concrete - simplified stress-block for deck moment capacity
IS 1786	2008	Reinforcement steel properties
IS 1893 (Part 3)	2014	Earthquake resistant design
IS 2062	2011	Structural steel - yield and ultimate strength by grade

8.2 Analysis and Design Assumptions of This Version

Structural Analysis

- All girders are modelled as simply supported; continuous spans are not currently supported.
- A 3D grillage model (OSPGrillage) is used for load distribution. Grillage members carry composite section properties after the construction stage.
- The transverse member forces are computed using an approximate 2D frame analogy. For irregular or skewed geometries, a 3D FEM is recommended.
- Fixed bearing stiffness is modelled as $k = 1,000,000$ kN/m (virtually rigid); free/expansion bearing as $k = 100$ kN/m.
- Construction stage sequence analysis is approximate. Detailed staged analysis should be performed for long-term deflection checks.

Composite Action

- Full shear connection is assumed at ULS with headed stud connectors designed per IRC 22:2015 Cl.606.
- Short-term composite section properties (modular ratio $n = E_s/E_{cm}$) are used for ULS checks.
- Long-term composite section properties (with creep-adjusted modular ratio) are used for SLS deflection and crack-width checks.
- Pre-composite stage: the steel girder alone resists all construction loads prior to concrete gaining strength.

Material Properties (IRC 22:2015 Annex III)

- Steel: $E_s = 200,000$ MPa, $G_s = 80,000$ MPa, $\nu = 0.30$, $\alpha = 11.7 \times 10^{-6}/^\circ\text{C}$ (grade-independent).
- Minimum structural concrete grade: M25.
- Default reinforcement grade: Fe500.

Partial Safety Factors (IRC 22:2015 Cl.601.4)

- γ_{m0} (steel yield, ULS) = 1.10
- γ_{m1} (steel ultimate, ULS) = 1.25

- Reinforcement (ULS) = 1.15
- Welds – shop: 1.25; field: 1.50
- Fatigue (γ_{mft}) = 1.35

Loading

- Dead load densities per IRC 6:2017 Cl.203: structural steel 78.5 kN/m^3 , concrete 25.0 kN/m^3 , bituminous wearing course 24.0 kN/m^3 .
- Multi-lane live load reduction factors per IRC 6:2017 Cl.204.4 Table 6A: 1st lane = 1.0, 2nd lane = 0.8, 3rd lane onwards = 0.4.
- Impact factor (dynamic load allowance) computed from span per IRC 6:2017 Cl.208.2/208.3.
- Wind load applied as transverse, longitudinal, and vertical components per IRC 6:2017 Cl.209.3.3–209.3.5.

Serviceability Limits

- Deflection limits (IRC 22:2015 Cl.604.3.2): live load + impact $\leq L/800$; total $\leq L/600$.
- SLS stress limits: concrete $\sigma_c \leq 0.48f_{ck}$; rebar $\sigma_s \leq 0.80f_{yk}$; steel $f_e \leq 0.9f_y$.
- Permissible crack width: $w_k \leq 0.3 \text{ mm}$ (bridge deck, exposure class XS2/XD2 per IRC 112:2020 Cl.12.3.2).

Fatigue

- Reference fatigue life: $N_{sc} = 2 \times 10^6$ cycles (IRC 22:2015 Cl.605).
- Constant stress range is assumed. A thickness correction factor μ_r is applied for plate thickness $> 25 \text{ mm}$.

Stiffener Design

- Intermediate transverse stiffener and bearing stiffener design follows IS 800:2007 Cl.8.7.2/8.7.3 and IRC 24:2010 Cl.509.7.2/509.7.3.

8.3 Known Limitations of This Version

- Substructure (piers, pile caps, foundations) and bearing design are not included.
- Splice connection design is not implemented.
- Skew angle > 15 degrees requires independent manual analysis (IRC 24 Cl. 504.8).
- Construction stage sequence analysis is approximate; detailed staged analysis should be performed for long-term deflection checks.
- The grillage analysis assumes simply supported boundary conditions; continuous spans are not currently supported.

References

1. IRC 5 (2024) — *Standard Specifications and Code of Practice for Road Bridges, Section I: General Features of Design.*
2. IRC 6 (2017) — *Standard Specifications and Code of Practice for Road Bridges, Section II: Loads and Load Combinations.*
3. IRC 22 (2014) — *Standard Specifications and Code of Practice for Road Bridges, Section VI: Composite Construction (Limit State Design).*
4. IRC 24 (2010) — *Standard Specifications and Code of Practice for Road Bridges, Section V: Steel Road Bridges (Limit State Method).*
5. IRC 112 (2011) — *Code of Practice for Concrete Road Bridges.*
6. IRC SP 114 (2018) — *Guidelines for Seismic Design of Road Bridges.*
7. IS 800 (2007) — *Indian Standard: General Construction in Steel — Code of Practice.*
8. IS 2062 (2011) — *Hot Rolled Medium and High Tensile Structural Steel — Specification.*
9. Subramanian, N. (2008). *Design of Steel Structures.* Oxford University Press.
10. Steel-INS DAG Teaching Resource Materials. <https://www.steel-insdag.org>